



ORTA DOĞU TEKNİK ÜNİVERSİTESİ
MIDDLE EAST TECHNICAL UNIVERSITY

Japan's 2011 Earthquake: Vulnerability, Social and Systemic Risk Analysis

Final Project

Ayşe Büşra Aktaş	2561017
Elif Büşra Algül	2428860
Hadiya Khan	2602084
Mürüvvet Eda Koyuncu	2561413
Zeynep Fenercioğlu	2561264

TABLE OF CONTENTS

1. ABSTRACT.....	3
2. INTRODUCTION.....	3
3. LITERATURE REVIEW.....	3
4. AIM AND OBJECTIVES.....	4
4.1 Aim.....	4
4.2 Objectives.....	5
5. METHODOLOGY.....	5
6. ANALYSIS.....	5
6.1 Exploratory Data Analysis.....	5
6.2 Inferential Analysis and Predictive Modeling.....	9
7. RESULTS & DISCUSSION.....	20
8. CONCLUSION & FUTURE WORK.....	21
9. REFERENCES/BIBLIOGRAPHY.....	22

1. ABSTRACT

This research paper presents an in-depth statistical assessment of Japan's seismic activity by using a dataset of 9,636 earthquake events of the United States Geological Survey (USGS), dating between 2005 and 2025. The research also transcends historical accounts by employing empirical techniques, such as correlation matrices, time-location study, and spatial mapping, to describe the behavior of seismic events in one of the most active tectonic areas in the world.

2. INTRODUCTION

Japan is one of the most active areas of seismic activity in the world, being placed along the Pacific Ring of Fire with several large tectonic plates converging. This topography brings about a high rate of earthquakes and, in some cases, disastrous therefore, seismic risk becomes a key issue to the infrastructure, community, and disaster preparedness in the country. The Great East Japan earthquake of 2011 showed the capacity of one extreme event to cause compound disasters, such as destructive waves of a tsunami, and other technological disasters, including the Fukushima Daiichi nuclear accident. The plans of natural hazards, structural vulnerabilities, and social outcomes became as interconnected as possible, which also showed the necessity of constant and data-driven evaluation of seismic activity in Japan.

Since the earthquake activity may change over time because of some change in tectonic stress, faulting, or the measurement practice, the issue of whether earthquake magnitudes have any long-term trends is becoming very important. These trends cannot be identified without a stringent use of statistical tools as opposed to anecdotal perception or a single incident. Equally, the spatial distribution of the earthquakes is a very important element of the seismic hazard evaluation. The earthquakes do not usually happen at random; they usually happen in clusters along the active faults or the plate boundaries. Determining the presence of strong spatial clustering in earthquakes and determining the location of the earthquake hotspots will be of great value in terms of risk modeling, land-use planning, and the design of resilient infrastructure.

This paper will address two key questions to gain a deeper insight into Japan's seismic landscape: (1) do earthquake magnitudes exhibit statistically significant increasing or decreasing trend with time, and (2) do earthquakes in Japan exhibit spatial clustering and, in that case, where are the major seismic hotspots? Based on a detailed database of earthquake events provided by the United States Geological Survey (USGS), the paper implements both the time trend and spatial pattern detection methods, which involve the use of statistical tools. This study brings relevant and multidimensional information on seismic risk in Japan, one of the most seismically active regions on Earth, through the combination of quantitative results with the generalized scientific knowledge of the geophysical environment of the country.

3. LITERATURE REVIEW

The 2011 Great East Japan earthquake and tsunami have been broadly described as a giant composite disaster due to the combination of extreme naturally caused and technology-related hazards. The earthquake of 9.0 under the sea caused a disastrous tsunami flooding, with run-up heights reaching more than 39 meters in some places and led to a cascade of technological malfunctions that resulted in the Fukushima Daiichi nuclear accident. The extent of this cascading event and the impact it has on disaster planning and risk management are continuously highlighted in the literature in the areas of disaster science, demography, engineering, and public health (Suppasri et al., 2013).

The nature of the physical processes, such as coastal overtopping, river-dike failures, and large-scale crustal subsidence are also a complex process and are emphasized in descriptive and hazard-oriented studies. These

results demonstrate that current coastal defenses and design assumptions were inappropriate in extreme and low-probability events. Quantitative measures state colossal human and economic costs, tens of thousands of deaths or disappeared people, mass destruction of houses, and economic losses of trillions of yen (Suppasri et al., 2013).

One significant demographic trend recorded in the literature is the over-representation of older adults in mortality with most of them aged 60 or older being victims in some prefectures. This has made age-related vulnerability a major concept in disaster studies, which has supported the necessity of special protection of aging populations (Hikichi et al., 2017).

The Fukushima Daiichi nuclear accident is often examined as a major systemic failure: tsunami waves were much higher than the protective design of the plant, emergency power and cooling systems were lost, which led to damage to the core. This case underlines mutual interdependence between the extremes of natural hazards and technological infrastructure risks as examples of the necessities of an integrated, interdisciplinary approach toward disaster preparedness (Kurokawa et al., 2012).

One of the most influential research directions is based on the Iwanuma Study, a rare longitudinal dataset that has pre-disaster baseline measurements (seven months before the occurrence). Results of this cohort indicate significant long-term non-economic impacts. Research findings describe an acceleration of cognitive deterioration in older adults, which is observed in 2.5 years, and unremitting depressive symptoms, which are closely related to housing devastation, displacement, and relocation conditions. These results broaden the scope of disaster literature to include both cognitive and mental health development over time, rather than the short-term perspective, which centers on PTSD (Kino et al., 2020).

The protective effects of social capital are another frequent theme that occurred in recent research. Strong pre-disaster community cohesion has been demonstrated to minimize severe symptoms of PTSD and cushion the negative cognitive and mental consequences. Furthermore, the composition of post-disaster relocation, especially its ability to maintain or break current social networks, turns out to be a major factor in recovery (Hikichi et al., 2017). It seems to be protective in the case of group relocation that preserves community relationships and desocializing in the case of random individual distribution that leads to higher levels of social isolation and depressive symptoms (Sasaki et al., 2019). The active social capital is of particular importance in maintaining cognitive health in older adults who engage in social activities (i.e., active) (Hikichi et al., 2017).

Taken in aggregate, the literature shows that the effects of the 2011 disaster were conditioned by the intensity of the physical hazards, as well as demographic composition, vulnerability of the infrastructure, and the social processes on the community level (Hikichi et al., 2017; Suppasri et al., 2013). These lessons imply the significance of interdisciplinary models which combine engineering, social cohesion, and the view of public health in the evaluation of disaster risk and recovery policy design (Kurokawa et al., 2012).

4. AIM AND OBJECTIVES

4.1 Aim

This research is expected to examine the temporal and spatial changes in earthquake occurrences in Japan through the application of sound statistical methods, to establish whether there is any significant change in the magnitude of the earthquakes over time, and to establish the locations of the major seismic hot spots. By adopting this two-way strategy, the study aims to improve the knowledge on the seismic risk scenario in Japan and contribute to evidence-based disaster preparedness and planning.

4.2 Objectives

- To ascertain whether the magnitude of earthquakes in Japan has a statistically significant increasing or decreasing trend with time, using the right statistical tests like linear regression or Mann-Kendall trend test.
- To test the presence of statistically significant spatial clustering of earthquakes in Japan with a spatial statistics test, e.g., Moran I, Ripley K-function or a hot spot test.
- To locate and delineate the major seismic hotspots in Japan, to pinpoint areas where there is clustering of seismic activity.
- In order to put results into perspective with the rest of the geophysical environment in Japan, to find correlations with tectonic plate boundaries, fault systems, and previous seismic activities.
- To illustrate the significance of statistical testing in earthquake studies, it is important to underscore the fact that empirical and data-driven approaches to earthquake behavior offer credible explanations of the data and guidance to policy and risk reduction.
- To add an interdisciplinary perspective on seismic risk through a combination of statistical results and knowledge obtained in disaster science, infrastructure vulnerability, and impacts on society due to large-scale earthquakes experienced in Japan.

5. METHODOLOGY

The data that was employed in this paper was taken from the United States Geological Survey (USGS), which is a globally acknowledged and scientifically credible body that continuously observes seismic movements with a system of sensors and satellite systems. We have chosen the dataset due to its consistency and quality, and globally standard measurements of the earthquake properties, i.e., magnitude, depth, geographic location, and the time when the event took place. We have a dataset of 9, 636 earthquakes that have occurred in and around Japan in a broad time range up to 2025. We selected this dataset since it is comprehensive, publicly available, morally acceptable, and large enough to offer significant statistical power. Also, its time sensitivity and stability make it especially well-suited to the statistical trend tests, including the Mann-Kendall test or linear regression.

6. ANALYSIS

6.1 Exploratory Data Analysis

6.1.1 Description of Variables

The variable “Time” indicates the date and time of the time the earthquake happened. “Latitude” and “Longitude” present the geographical location of the epicenter of the earthquake. “Depth” indicates the depth of the hypocenter of the earthquake beneath the surface of the earth which is in kilometers. “Magnitude” explains the magnitude of the reported size or the amount of energy that was emitted by the seismic event. “Magnitude type” is the scale/calculation technique of measuring the magnitude of the earthquake. “Gap”, the largest angle between the seismic stations around the epicenter, is known as the azimuthal gap, and it is typically a measure of quality of location. “Place” is a textual description of the location of the earthquake, usually a reference to the closest city or to a geographic area. “Mw_est” is the estimated moment magnitude (Mw) that provides a more standardized measure of earthquake size that allows comparisons with other events. “Location Source” specifies the source of data or network of seismics, which located the geographic position of an earthquake, and “Magnitude Source” is the source which measured or reported the magnitude of an earthquake.

6.1.2 Data Description

From Table 1, the descriptive statistics reveal that the earthquakes are spatially clustered along the eastern part of Japan, and the latitude and the longitude have low variation. The depths of earthquakes show a broad distribution and a skewed right, demonstrating the existence of shallow and deep earthquakes. The magnitudes are largely moderate and narrowly focused around their mean values, whereas the gap values exhibit moderate variation, implying that there is disparity in the coverage of the stations in events.

Variable	Mean	Median	Std. Dev.	Min	Max
Latitude	36.764	37.088	3.041	30.001	44.951
Longitude	141.326	141.910	2.936	129.037	145.990
Depth (km)	44.439	35.000	57.573	0.600	644.879
Gap (°)	103.142	113.200	36.349	8	277
Mw_est	5.061	5.000	0.307	4.5	9.1

Table 1: Basic Statistics

In Table 2, the correlation matrix reveals that there is a moderate positive relationship between latitude and longitude (0.53), implying that the locations of earthquakes in Japan have a consistent geographical distribution around the archipelago. Depth has a weak and negatively related to longitude ($r = -0.25$) and an insignificant correlation with latitude ($r = -0.02$), which indicates that earthquake depth depends largely on the geographic location. Magnitude (Mw est) has no significant relationship with latitude, longitude, or depth, meaning that there is no favorable occurrence of the larger earthquakes in the study area in terms of location or depth. Nonetheless, Mw est has a moderate negative correlation with gap (-0.50), which means that large earthquakes are more likely to have a smaller azimuthal gap and hence a higher coverage of the stations.

Variable	latitude	longitude	depth	gap	Mw_est
latitude	1.0000	0.5270	-0.0167	0.0762	0.0525
longitude	0.5270	1.0000	-0.2473	0.2061	0.0269
depth	-0.0167	-0.2473	1.0000	-0.1607	-0.0259
gap	0.0762	0.2061	-0.1607	1.0000	-0.5029
Mw_est	0.0525	0.0269	-0.0259	-0.5029	1.0000

Table 2: Correlation Matrix

Figure 1 indicates that the latitude distribution ranges between about 30°N and 45°N with a clear concentration of earthquakes in the range of 35°N and 40°N. Distribution is unimodal with an intense peak of 37°N. Figure 2 shows that longitude coverage is between 129°E to about 145°E, and most of the earthquakes are located at 140°E to 145°E. This is highly left-skewed, and the clustering is sharp along the eastern side of Japan. The depth distribution in Figure 3 is greatly skewed to the right, with the greatest number of earthquakes having depths of 0 km, then, within the 0 km to about 50-60 km range. The frequency also goes down sharply with depth, and only a few events take place more than 100 km deep. Figure 4 shows that the distribution of magnitude is heavily right-skewed; most of the earthquakes are concentrated in the range of magnitude 4.0 to 5.5. The frequency decreases sharply with increasing magnitude, producing a relatively small number of events with magnitude greater than 6.0 and a very small number of events with magnitude greater than 7.0.

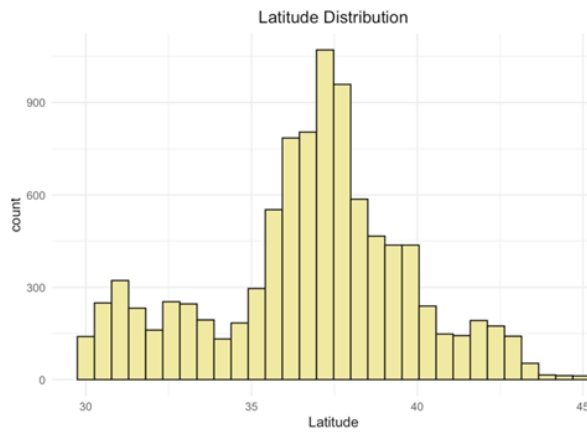


Figure 1: Latitude Distribution

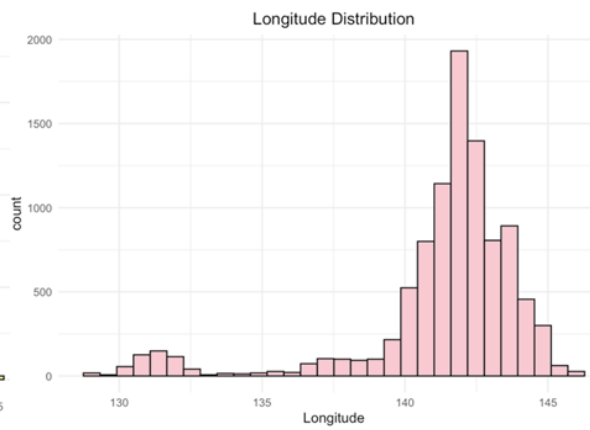


Figure 2: Longitude Distribution

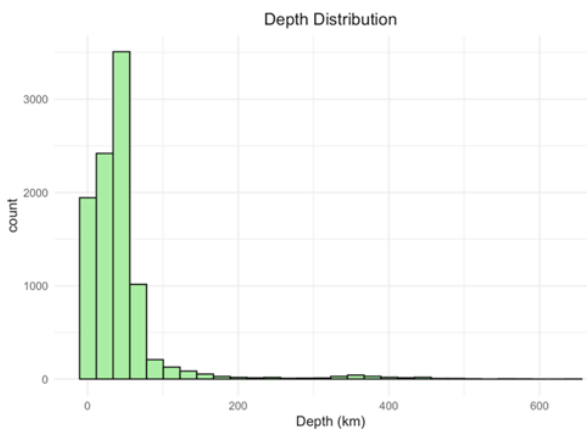


Figure 3: Depth Distribution

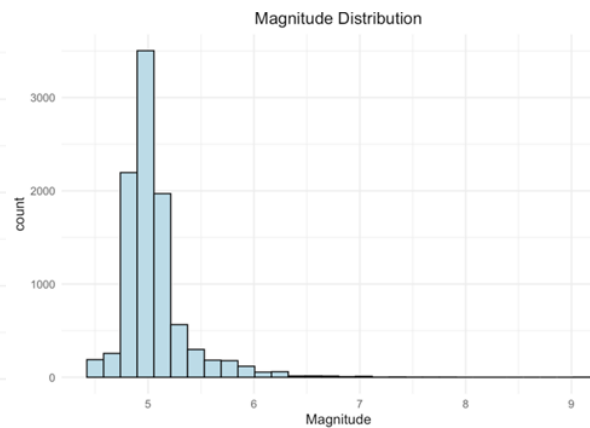


Figure 4: Magnitude Distribution

The bar chart, in Figure 5, indicates that there is a huge percentage of earthquakes that belong to the light and moderate magnitude categories. The largest category is light earthquakes, and the second category is moderate earthquakes, which means that the most frequent seismic activity in the data is the low-to-moderate magnitude earthquakes. Compared to this, strong earthquakes are quite infrequent, and the major earthquakes happen in very few cases. This distribution is the characteristic seismic distribution whereby even small earthquakes are much more common than large and destructive earthquakes.

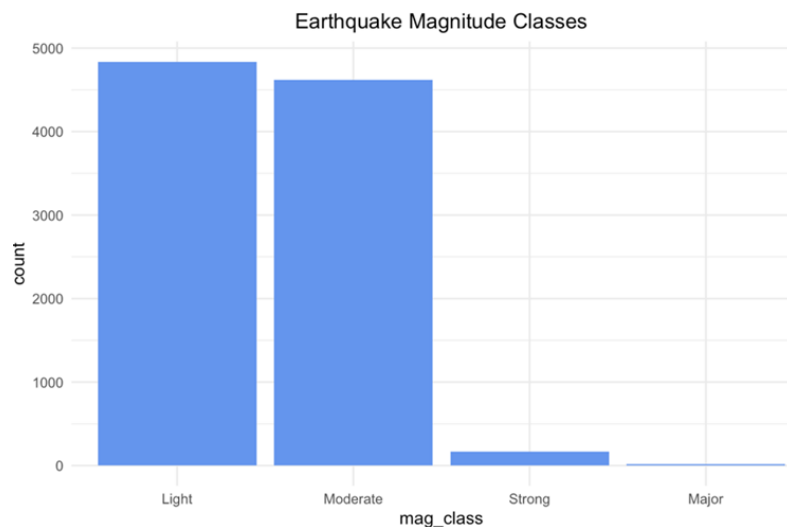


Figure 5: Earthquake Magnitude Classes

In Figure 6, the storyline reveals that the magnitude of earthquakes does not have a high linear correlation with depth. Most of the events take place at relatively shallow depths (under about 100 km), and the magnitude of most of the events is concentrated in the range of 4.5 to 5.5. Although deeper earthquakes (greater than 300 km) are not as common, they may also attain moderate magnitudes comparable to shallow ones. The trend line is smooth and shows that there is a weak and non-linear change in magnitude with increasing depth, indicating that depth alone is not a dominant determinant of earthquake magnitude.

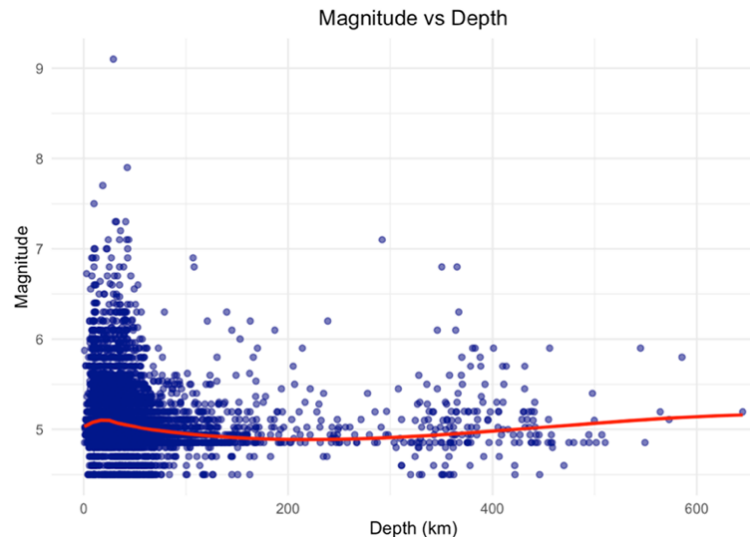


Figure 6: Magnitude vs Depth Scatter Plot

The number of earthquakes is quite low and steady between 2005 and 2010 which can be seen from Figure 7, followed by a steep rise in 2011, then the values drop once again and continue to oscillate at moderately high values until 2012 to 2025. All in all, the annual numbers demonstrate the non-uniform fluctuation, having one obvious peak year with more stable activity.

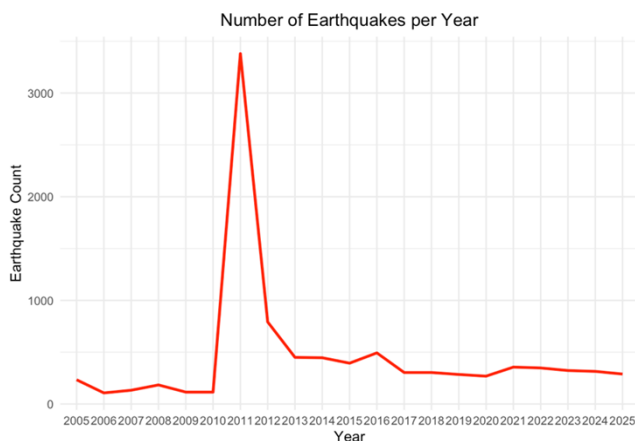


Figure 7: Number of Earthquakes per Year

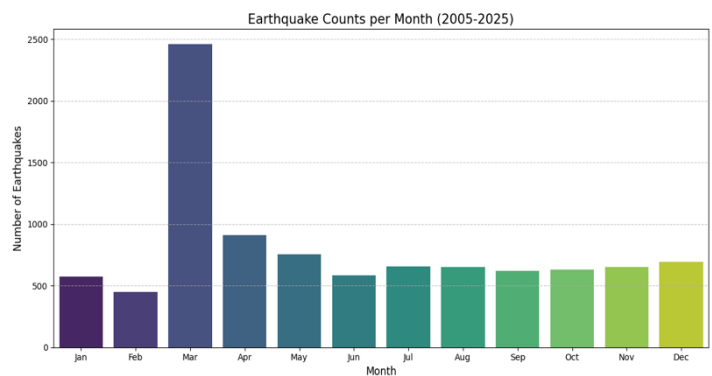


Figure 8: Number of Earthquakes per month

The earthquake epicenter map shows that there is a clear spatial concentration of the events in the eastern side of Japan, and more so in the offshore of Honshu, which can be seen from Figure 9. The western part of Japan has significantly less seismic activity, but one can see secondary clusters in the south and north. Such a pattern shows that there is no even distribution of earthquakes in Japan, and they are concentrated in certain tectonically active areas.

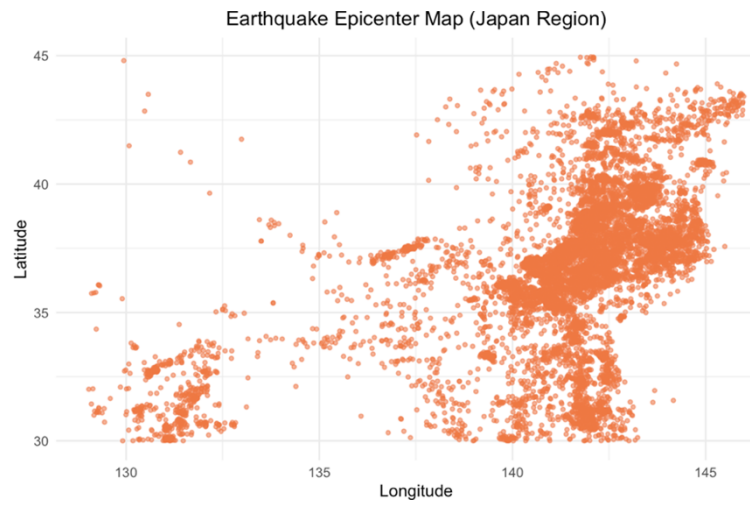


Figure 9: Earthquake Epicenter Map

Initial overview of our data, and according to the above map, hotspots with recurring seismic activity exist. Below, in Table 3, is a list of places with the most seismic activity in Japan:

Place	No. of seismic activity
Izu Islands, Japan region	876
off the east coast of Honshu, Japan	181
near the east coast of Honshu, Japan	80
Hokkaido, Japan region	16
Eastern Honshu, Japan	14
238 km ESE of Ishinomaki, Japan	14
233 km ESE of Ishinomaki, Japan	10
236 km ESE of Ishinomaki, Japan	10
Kyushu, Japan	9
119 km E of Miyako, Japan	9

Table 3: Top hotspots of seismic activity

6.2 Inferential Analysis and Predictive Modeling

Analysis of magnitude sources

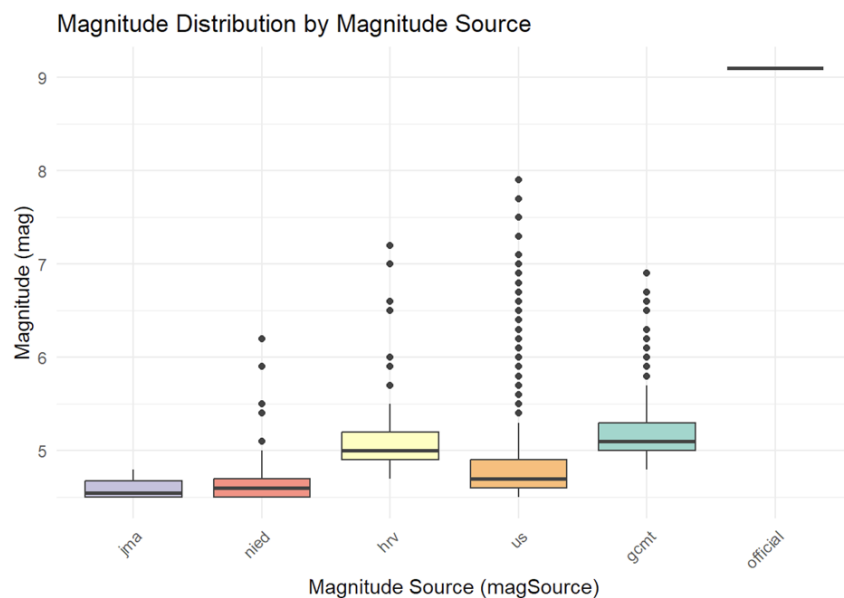


Figure 10: Analysis of magnitude sources: Distribution of magnitudes for different sources.

MagSource	Count	Mean_Mag	Median_Mag	SD_Mag	Min_Mag	Max_Mag
Official	1	9.1	9.1	NA	9.1	9.1
gcmt	497	5.16	5.1	0.303	4.8	6.9
hrv	114	5.12	5	0.439	4.7	7.2
us	8664	4.79	4.7	0.366	4.5	7.9
nied	354	4.66	4.6	0.194	4.5	6.2
jma	6	4.6	4.55	0.126	4.5	4.8

Table 4: Summary statistics for the distribution of magnitudes for different sources.

The plot reveals that various magnitude sources report various magnitudes. The example is that the magnitude of the earthquake reported by the USGS National Earthquake Information Center (NEIC) is lower than that reported by Hrv (Harvard Moment Tensor Project). Critical lessons learnt from these graphs are:

- The reason is that HRV and GCMT are concerned with reporting stronger and significant earthquakes around the world with magnitudes greater than 5 or 5.5, hence the changes in distribution towards a larger magnitude.
- USGS is the main high-volume source that has reported magnitudes that are spread broadly.
- HRV, US, and GCMT are mildly right skewed and outliers at larger magnitudes.
- The scope of events that JMA and NEID report is smaller.
- The largest outlier (mag = 9.1), which is provided by the official, is based on one data point.

Summary of Seismic Magnitude Analysis

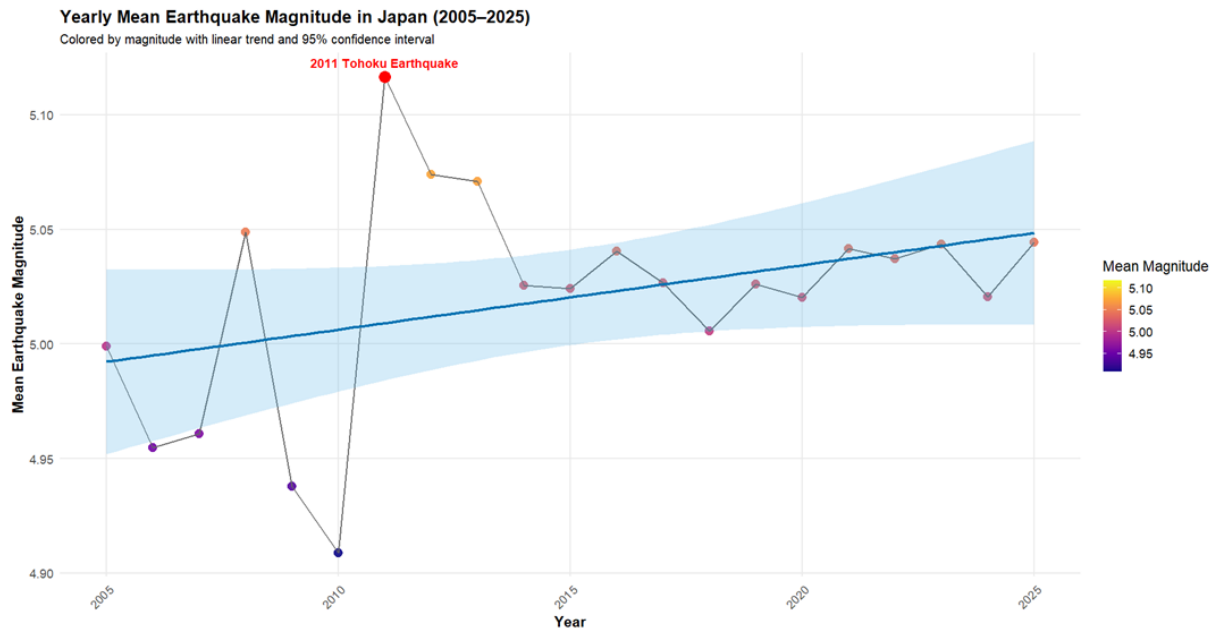


Figure 11: Yearly Mean Earthquake Magnitude in Japan

In the statistical analysis of the seismic activities in Japan, it is seen that the long-term stability is quite high, with an annual mean magnitude of the yearly events always centered around 5.0. Linear regression indicates a statistically significant negative temporal trend ($p < 0.001$), however, the average change in slope of the regression is very small, close to -0.003 , which corresponds to a magnitude of units per year. The explanatory power of the model is insignificant ($R^2 \approx 0.002$), a result supported by the Mann-Kendall test, which was able to identify a weak monotonic decreasing trend ($\tau = -0.0708$, $p < 0.001$).

The statistics underscore the fact that the 2011 Great East Japan Earthquake was such an isolated event, which was very unique and temporary in the sense that it increased the annual mean without altering long-term systematic behavior. Finally, the findings indicate that, although the downward trend can be statistically identified, the change is too insignificant to possess either practical or geophysical value, indicating that Japan does not experience any change in the seismic energy release over the years.

Summary of Magnitude Differences Across Depth Categories

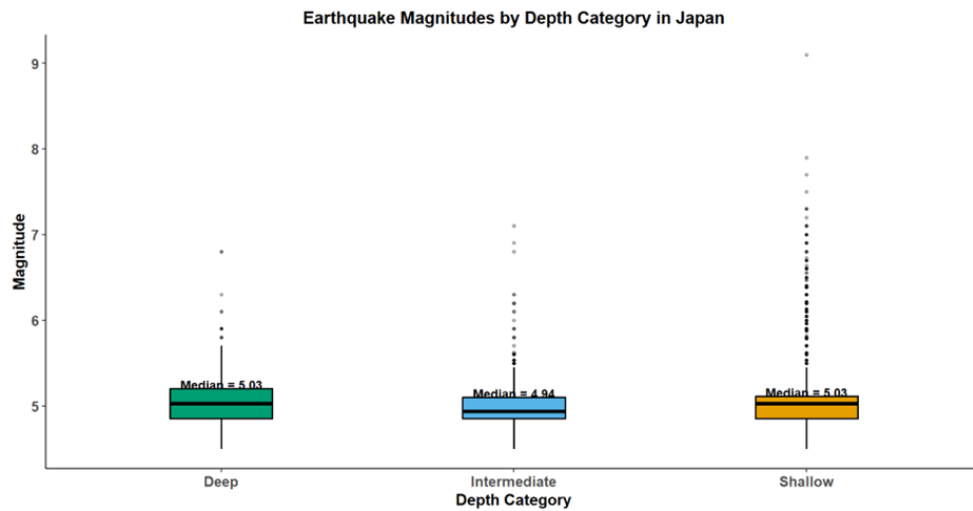


Figure 12: Earthquake Magnitudes by Depth Category in Japan

The analysis confirms that earthquake magnitudes in Japan differ significantly based on depth categories. While shallow and deep earthquakes share a median magnitude of 5.03, intermediate-depth events exhibit a lower median of 4.94. The Kruskal–Wallis rank-sum test establishes that these differences are highly statistically significant ($\chi^2 = 42.499$, $df = 2$, $p = 5.907 \times 10^{-10}$), proving that depth is a statistically significant factor in seismic size.

Post-hoc pairwise Wilcoxon tests with Bonferroni correction reveal that this global significance is primarily driven by the contrast between shallow and deep earthquakes ($p \approx 1.8 \times 10^{-10}$), whereas intermediate events do not show significant differences from either group. Despite this robust statistical result, the small effect size ($\epsilon^2 \approx 0.004$) indicates that depth explains only a minimal proportion of overall magnitude variability.

The boxplot visualization confirms these findings, showing that shallow earthquakes possess a wider range of extreme values and the highest observed magnitudes. This aligns with seismological theory, where the brittle nature of shallow crustal environments allows for greater stress accumulation and energy release compared to the more ductile conditions found at greater depths.

Investigation of seismic activity leading to the great 2011 disaster. Were there any abnormalities indicating a looming disaster?

The 11th of March, 2011 Tohoku Earthquake and the secondary Tsunami disaster is no doubt a *Black Swan event* that completely shook Japan. A powerful earthquake off the coast of Honshu triggered a Tsunami that took at least 20,000 lives and devastated the economy.

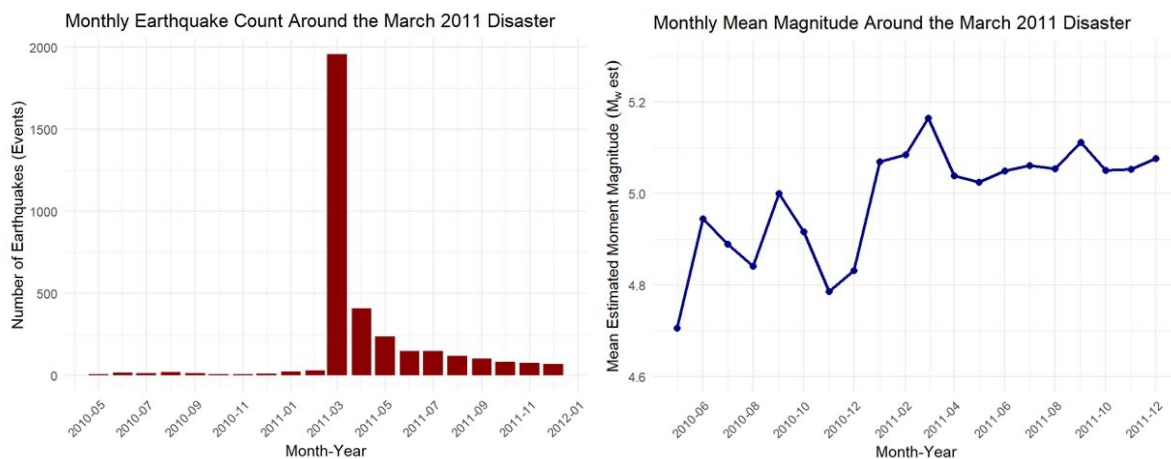


Figure 13: Summary statistics for distribution of magnitudes for different sources.

Mean monthly magnitude = 4.979

Mean monthly event count = 12.3

Visualizing the number of seismic events by months around March 2011 puts into perspective the absolute unexpected chaos of this disaster. The number of seismic events in March 2011 skyrocketed and despite reducing in further months, they still were high enough compared to overall mean event count of 12.3. Further zooming into the actual counts shows exactly how drastically removed the event was from normal.

Month/Year	Event count	Mean Magnitude (M_{est})
12-2010	8	4.83
01-2011	22	5.07
02-2011	28	5.08
03-2011	1955	5.16
04-2011	407	5.04
05-2011	236	5.02

Table 5: Monthly magnitudes leading up to the Tohoku disaster.

Let's map the seismic activity in the year 2011 to see what areas were actually affected.

Geographical Distribution of Earthquakes in March 2011

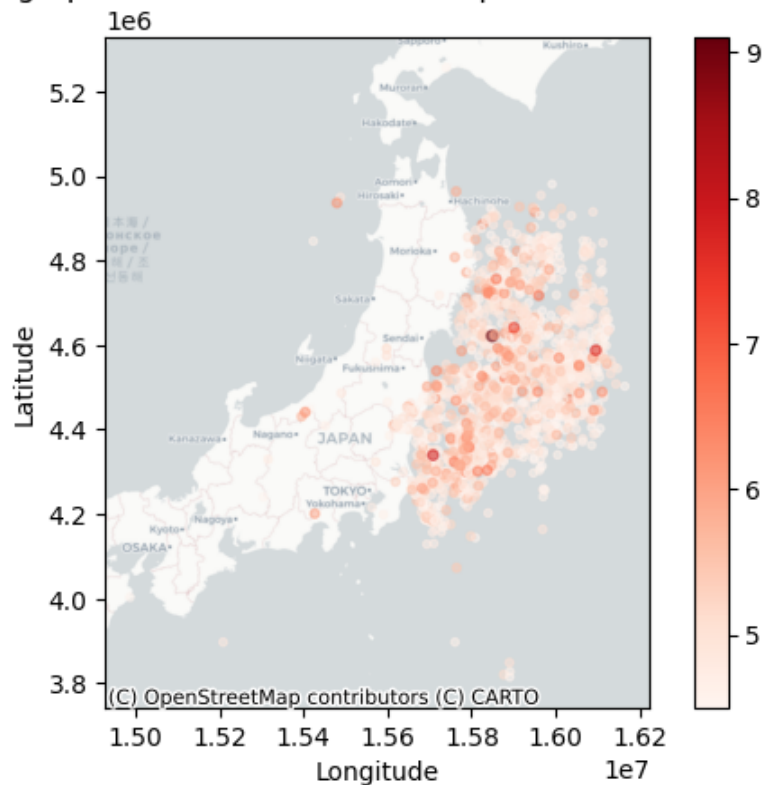


Figure 14: Map of earthquake events in March 2011.

This map provides a good visual of the events during the most significant geological event in modern times. The dark red points indicate the highest-magnitude events. The cluster of dots follows the Japan Trench where the Pacific Plate is sliding beneath the Okhotsk Plate. The activity was concentrated relatively close to the shore, but the sheer intensity and volume of shocks reverberated throughout the region. Aftershocks continued even after the main event making rescue and recovery efforts even more difficult.

- Concluding the events of 2011 great disaster we have the following key findings:
- The slight increase of events reveals a subtle trend before indicating that the zone was entering a state of heightened instability months before the main rupture.
- The tight clustering of points (Figure 18.) confirms the concentration of stress in that area. Because of proximity to the shore, energy release had a direct impact on land which is why aftershocks (magnitude 5-6) remained dangerous.

Scientists did not believe that the Japan Trench was capable of a 9.0+ event. The expected magnitude was around 8-8.5. The immense rupture and the resulting tsunami truly defied expectations.

Economic losses and fatalities for major events

One of the most primary concerns when analyzing earthquake data is to examine its losses, both economic and mortalities. We first filter our data for major seismic events and then merge it with data from EM-DAT (The International Disaster Database) and NOAA/NCEI (Global Historical Disaster Database). The link was based on time and location of the event.

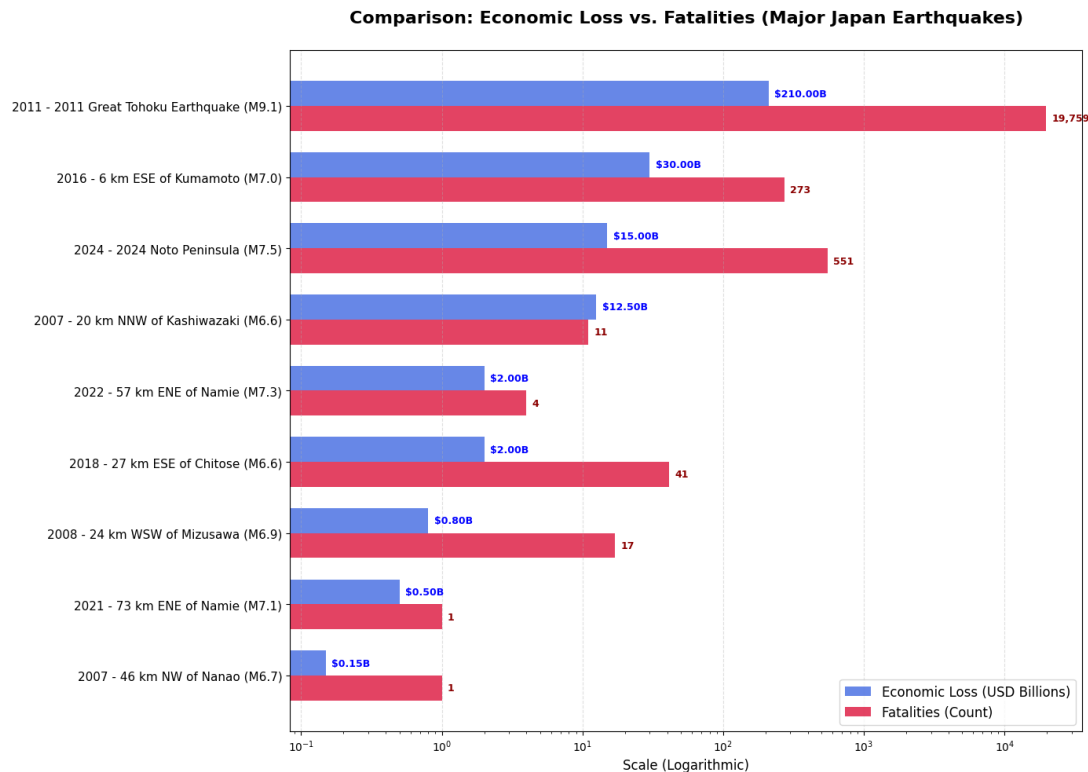


Figure 15: Comparison of economic losses and fatalities.

The above figure puts into perspective the losses due to each major event in Japan. The scale is log transformed so it's easier to see extreme losses of the Tohoku event along with others in one frame.

- **The "Tōhoku Gap":** The gap between the top event and the second (Kumamoto) is vast. Tōhoku was roughly 7 times more expensive and caused 72 times more fatalities than the next most destructive event in this 20-year window.
- **Infrastructure Sensitivity:** The 2007 Kashiwazaki event (6.6M) appears much higher in the economic chart than in the fatality chart. This highlights Japan's vulnerability in specific industrial zones where even "mid-sized" earthquakes can trigger massive financial losses.
- **Modern Resilience:** More recent events like the 2022 Namie (7.3M) show high magnitude but significantly lower fatalities and costs compared to older events of similar size, likely reflecting continuous improvements in early warning systems and construction standards.

Geophysical analysis (Tectonic plates and faults)

To further dive into the tectonic study of Japan, we can start by visualizing the tectonic zones. This will help us to find the seismic risks associated with different areas in Japan.

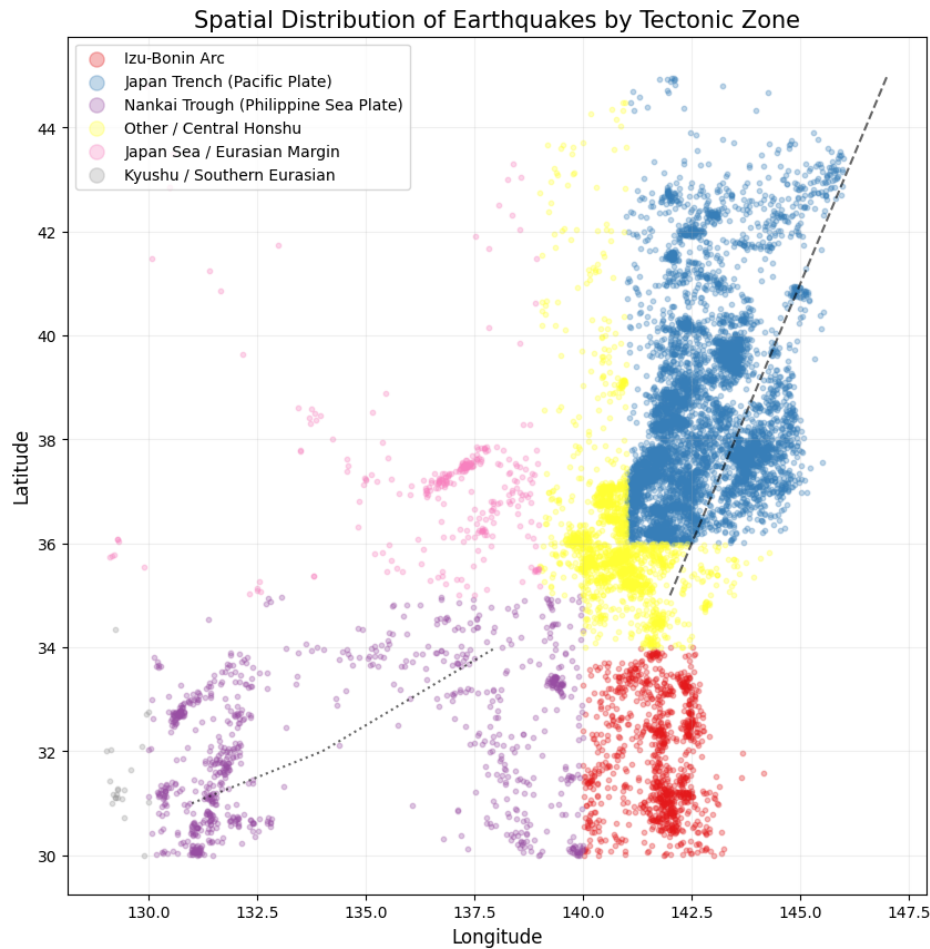


Figure 16: Spatial distribution of earthquakes by tectonic zones.

Tectonic Zone	Primary Plate Interaction	Total Economic Loss	Total Fatalities
Japan Trench	Pacific -> North American	\$214.50 Billion	19,805
Nankai Trough	Philippine Sea -> Eurasian	\$30.00 Billion	273
Japan Sea Margin	Eurasian / Okhotsk Collision	\$27.65 Billion	563
Central Honshu	Intraplate (Crustal) Faults	\$0.80 Billion	17

Table 6: Total economic losses and fatalities due to each tectonic plate (Data from EM-DAT and NOAA/NCEI)

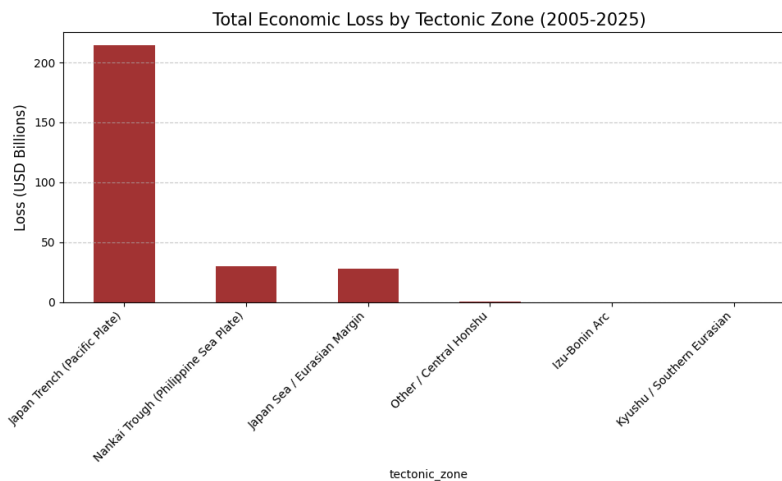


Figure 17: Economic loss by each tectonic zone (Data from EM-DAT and NOAA/NCEI)

Based on these above graphs, we can clearly conclude that the Japan Trench is the most active and violent area responsible for over 80% of total economic losses. Notice how the Japan Trench (indicated by blue dots in Figure 20) is exactly where the Tohoku 2011 earthquake happened. This could indicate that the Japan Trench is notorious for vicious seismic activity and hence this area should be in focus when laying out risk management plans.

We can further zoom in to visualize the cross-section profile of these plates and the depths of these earthquake events to arrive at feasible conclusions.

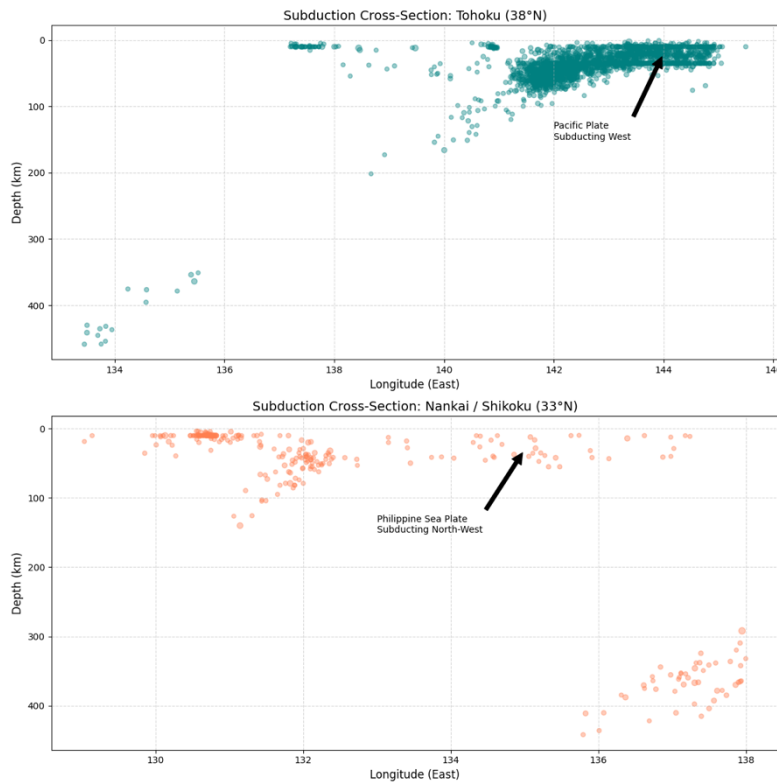


Figure 18: Subduction cross section graph.

The Subduction Cross-Section graph provides a look "inside" the Earth.

- Tohoku (38°N): You can clearly see the *Pacific Plate* dipping westward. The earthquakes follow a diagonal line (the Benioff Zone) that starts shallow at the trench (East) and reaches depths of over 200km as it moves under the mainland.
- Nankai (33°N): The *Philippine Sea Plate* has a shallower subduction angle. This is critical because a shallower angle means the "locked zone" (where the plates stick and then snap) covers a larger surface area closer to the coast, leading to higher tsunami risks for southern Japan.

Major secondary effect of earthquakes - Tsunamis

An immediate secondary effect of undersea earthquakes is a tsunami. Hence, we analyze our seismic data along with tsunami records of Japan taken from Japan Meteorological Agency (JMA) to identify what factors or thresholds trigger a tsunami. Below are our findings.

Total Tsunami Events Identified	8
Average Depth of Tsunami Quakes	27.62 km
Average Magnitude of Tsunami Quakes	7.40

Table 7: Tsunami findings. (Data from JMA)

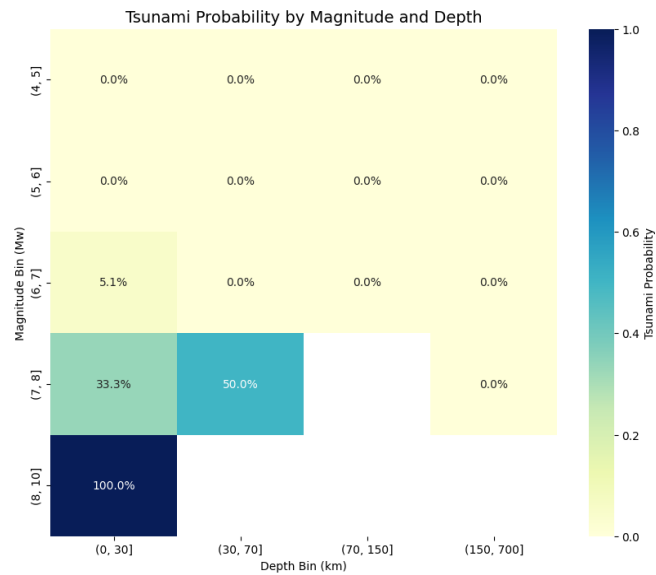


Figure 19: Tsunami probability matrix.

The above matrix plot shows the probability of occurrence of a tsunami based on a seismic activity's depth and magnitude. It is clearly seen that events with magnitude greater than 7 and depth between 0-70 km will likely cause a tsunami. Hence, preparedness for such events is vital.

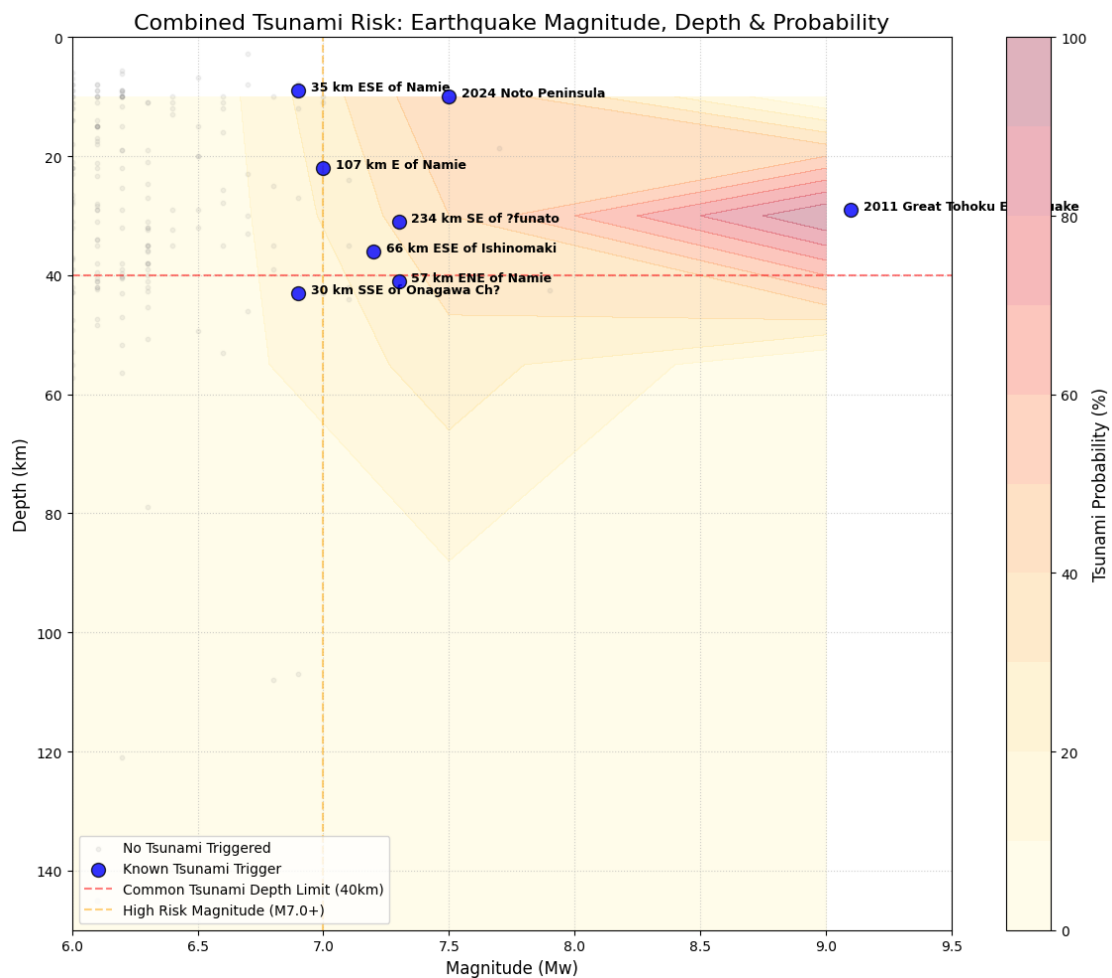


Figure 20: Plot of tsunami probability along with past events.

The above plot shows seismic events on a tsunami probability contour plot. The blue points indicate known events that triggered a tsunami. The immediate point we notice is the 2011 Great Tohoku disaster which is

a clear outlier. Notice how most tsunami generating events are concentrated in the upper right corner (magnitude >7 and depth >40). This validates our findings from the probability matrix we generated earlier.

Risk to critical infrastructure - Nuclear Power Plants (NPP) and Ports

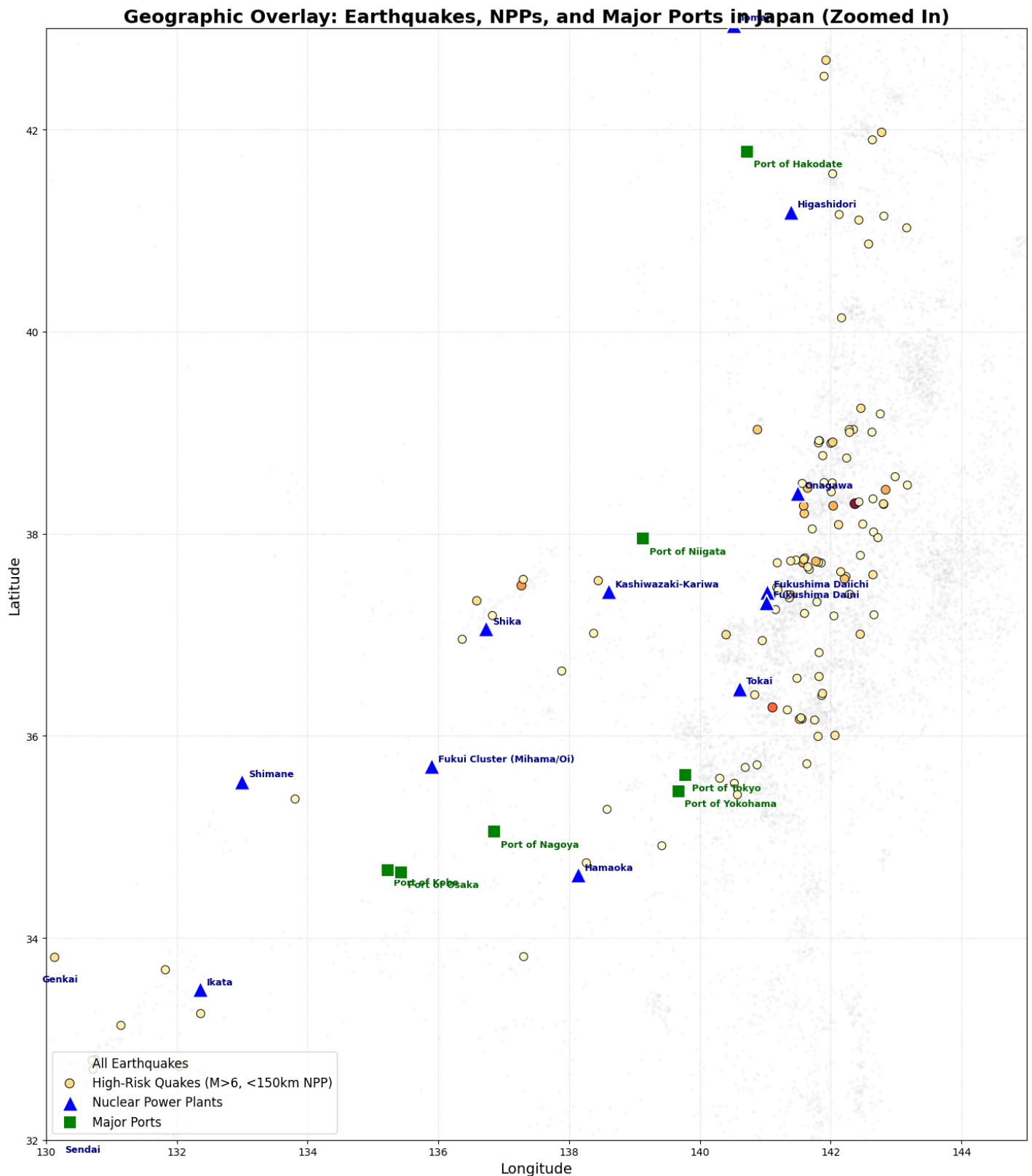


Figure 21: Map of high-risk events along with locations of critical infrastructure.

(NPP Data from NRA -Nuclear Regulation Authority, Japan, IAEA and WNA- World Nuclear Association)

(Port data from MLIT- Ministry of Land, Infrastructure, Transport and Tourism and Japan Port and Harbor Association)

This map shows locations of critical infrastructures namely NPP and ports along with high-risk events. As we can see, Japan has nuclear plants spread across its land. This makes high risk seismic events even more

dangerous. The areas in the top right of the map that have blue triangles (NPP's) surrounded by yellow points (high-risk quakes) are prone to very damaging events. These areas should have emergency measures in place in connection with the nuclear plants they house.

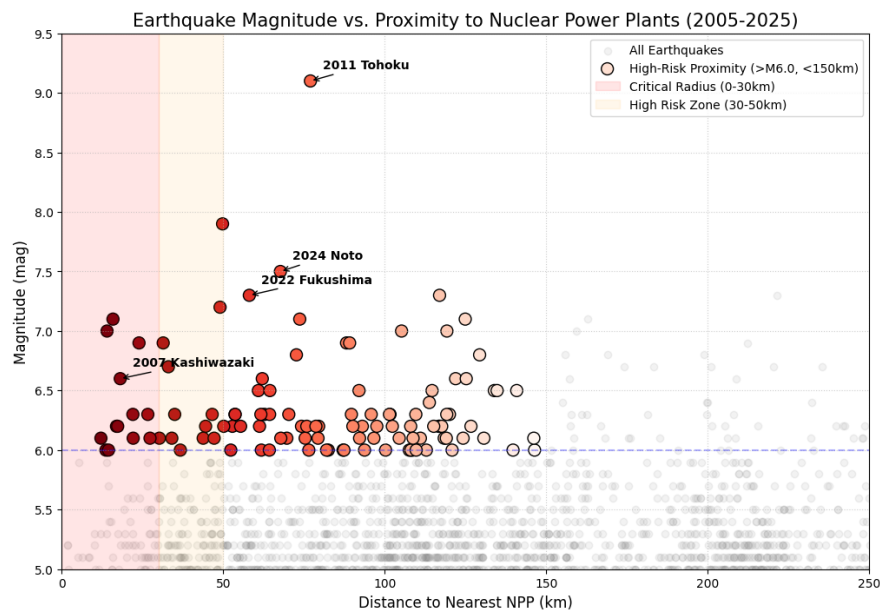


Figure 22: Earthquake events on magnitude vs distance to NPP.

To find the crucial thresholds of factors for an event to prove disastrous, we can visualize the magnitudes of seismic events along with their proximity to NPP. This above graph shows this along with critical radius bands highlighted in red. As we can see, the dark red dots are concentrated here which means the events within 30 km of a nuclear power plants and magnitude > 6 can prove to be most dangerous.

Predictive Risk Modelling

Based on the analysis we have done so far and the information gathered, we can generate a predictive model that specifies some characteristics of a future event. This may help stakeholders to narrow down their safety policies on target areas in high-risk zones, with critical infrastructures. Below are the predicted events our model has put forth:

Scenario	Magnitude	Depth	Distance	Estimated loss(\$B)
Nankai Megaquake (Offshore)	8.5	30	80	49.46
Tokyo Near-Field (Crustal)	7.2	10	20	3.39
Sea of Japan Shallow	6.8	10	30	1.65

Table 8: Forecasted future events.

Another tool that may help us to predict losses of seismic events based on their characteristics is to plot the predictive risk model we used before.

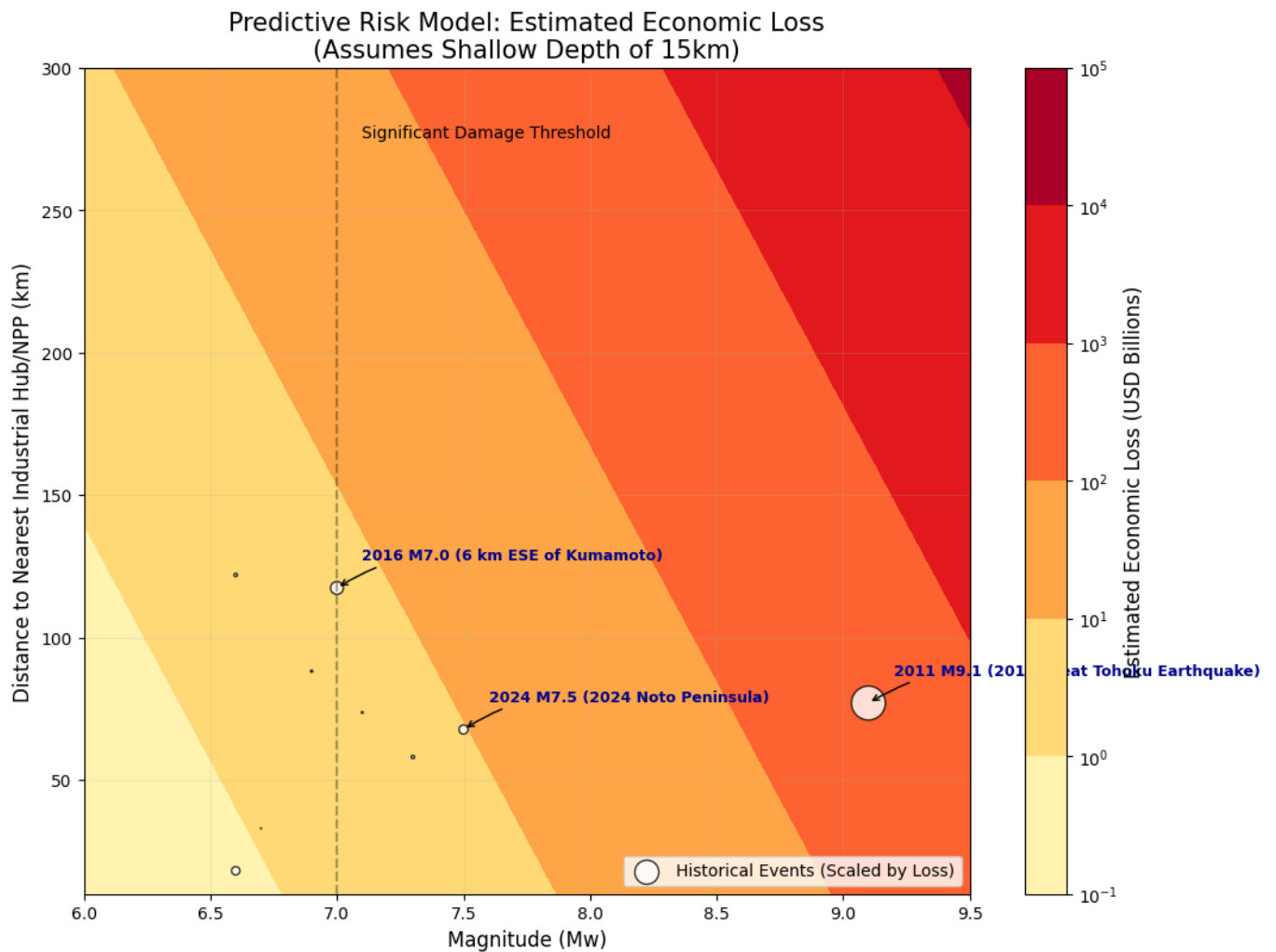


Figure 23: Predictive risk model with estimated loss.

This graph puts into perspective the cost of future disasters along with past major events. Notice how the 2011 Tohoku disaster falls in the dark orange band, indicating a loss of 100-1000 billion dollars. As expected, economic loss increases as proximity to NPP of a seismic event increase. This coupled with increasing magnitudes means that events can truly be very disastrous.

7. RESULTS & DISCUSSION

This paper presented a multi-layered, in-depth analysis of the activity of earthquakes in Japan integrating descriptive statistics, correlation analysis, time-related trends, spatial mapping, tectonic explanation, tsunami evaluation, and infrastructure risk assessment. The analysis was based on a complete dataset on earthquakes during the span of 2005 to 2025 to address the more complex aspects of the relationship between seismic parameters and geological structures as well as societal exposure than individual magnitude-based interpretations of the relationship.

The statistical summary indicated that most earthquakes in Japan occur at shallow to intermediate depths with most of them falling within the 4.5 and 5.5 magnitude. The proportion of extreme events is not huge, but the effect is excessive. The correlation analysis revealed that most seismic variables are connected to each other in a rather weak linear manner which displays a nonlinear and multi factor tectonic process that rules the behavior of earthquakes. The moderate negative relationship between magnitude and seismic gap, however, leads to the idea that bigger earthquakes are prone to be taking place in the highly active, well-instrumented tectonic regions as opposed to the regions that have fewer monitoring instruments. Also, the absence of a strong logical relation between depth and magnitude supports the finding that the earthquake size cannot be highly limited by the focal depth only. Long- and short-term seismic patterns were identified using temporal analysis. The sharp increase in the rate of earthquakes in March 2011 is an indicator of the short term and long-term sequence of aftershocks of the great Tohoku earthquake. This increase in the number of events notwithstanding, the average annual magnitude over the long-term was relatively constant which may reflect the fact that large earthquakes do not always enhance the average seismic energy release in the long run. This observation presents the need to evaluate frequency and intensity independently when evaluating seismic hazards.

Spatial visualizations indicated that earthquakes are highly clustered along major tectonic boundaries, especially the Japan trench, Ninkai trench and the Izu- Bonin arc. The areas agree with the known subduction areas where the Pacific and Philippine Sea plates collide with the Eurasian and North American plates. Subduction geometry was also well demonstrated on the epicenter density maps and the depth profiles across the sections with superficial seismicity observed on the trench progressing to more profound landward based earthquakes. These trends validate the leading role of plate tectonics in regulating the distribution of seismicity space and depth trends in Japan.

The combination of magnitude sources showed that there was small yet systemic differences in reporting agencies, which underline the need to consider the variation in measurements during the aggregation of seismic catalogs. Although these differences did not affect the trends at a significant level in the regions, they create uncertainty, which is even more important, when considering extreme events or conducting comparative studies across data sets.

One of the most important contributions of this work is the risk-oriented analysis. The earthquake-tsunami evaluation revealed that tsunamis are much more probable in earthquakes with magnitudes of more than Mw 7.0 and depths less than about 40 km. The outcome of this finding is that potential tsunami is controlled by the interaction of magnitude, depth, and offshore position but not magnitude. The possibility tsunami matrix and conjoined risk plots further indicated that several events that have happened in the past that have been devastating are within this high-risk parameter space and this proved why integrated hazard indicators are necessary. Studies based on infrastructural factors showed that numerous high-magnification earthquakes happened close to nuclear power plants and key ports in non-negligible numbers. The proximity-based risk assessment demonstrated that even moderate magnitude earthquakes can have serious secondary risks, given that the distance between them is in the critical range. These comparisons of losses and lives lost in tectonic zones indicated that the events relating to subduction, especially those occurring along the Japan Trench can be responsible for the vast majority of all the losses, with intraplate earthquakes, though more infrequent, still capable of causing severe local damages.

Lastly, the predictive loss modelling was a futuristic view in that it predicted the possible economic losses in probable scenarios of an earthquake. Such findings revealed just how fast the anticipation of losses was

increasing with the magnitude and the distance of the critical infrastructure, which emphasized the nonlinear nature of the seismic risk. These results highlight the significance of preparedness and mitigation measures that should consider tectonic context as well as infrastructure exposure. Summing it up, this project shows that the risk of earthquakes in Japan cannot be properly explained by using univariate analysis. Rather, a comprehensive system, which incorporates statistical characterization, spatial-tectonic setting, temporal behavior, and risk-based modeling, is necessary to give the actual complexity of seismic phenomena. The results strengthen the special seismic susceptibility of Japan as well as offer an analytical framework that may be extended to other tectonically active areas on Earth.

8. CONCLUSION & FUTURE WORK

Based on the multi-dimensional results of the current research study, there are several areas of advancement that can be made to both deepen the scientific and practical value of the analysis.

Although this research is a comprehensive exploratory and risk-based study, it can be extended in many ways to enhance the strength and breadth of findings. First, it would be possible to consider ground motion parameters which include peak ground acceleration (PGA) or the seismic intensity measures to have a more direct assessment of the potential of structural damage beyond magnitude-based interpretations.

Second, the incorporation of population density, land use, building typology and infrastructure vulnerability data would contribute greatly to economic loss and casualty estimation models. This integration would facilitate a better separation of seismic hazard and societal risk especially in the densely populated coastal areas.

Modeling could be used to predict future work with machine learning or probabilistic models to predict losses caused by earthquakes or likelihood of a tsunami based on multi-variable inputs, such as magnitude, depth, fault mechanism, tectonic setting and closeness to critical infrastructure. Besides that, time-series and point-process models, e.g., Epidemic-Type Aftershock Sequence (ETAS) model, could be used to describe the behavior of aftershocks and temporal clustering of aftershocks better after major seismic events.

Another extension is uncertainty quantification. Modeling uncertainties explicitly in the magnitude estimation, depth measurements, tsunami occurrence and economic loss data would better instill confidence in risk predictions and increase the interpretability of results among the decision-makers.

Lastly, the expansion of the framework to include real-time seismic data and the checking of the effective functionality of early warning systems would be a valuable source of information in disaster preparedness and mitigation. The developments may benefit the emergency response planning, infrastructure resilience plans and long-term policy moves in the earthquake prone areas like Japan directly.

9. REFERENCES/BIBLIOGRAPHY

Ahlenius, H. (2022, March 19). *World tectonic plates and boundaries*.

GitHub. <https://github.com/fraxen/tectonicplates>

EM-DAT. (2024). *EM-DAT | The international disasters database*.

Emdat.be. <https://www.emdat.be/>

Hikichi, H., Tsuboya, T., Aida, J., Matsuyama, Y., Kondo, K., Subramanian, S. V., &

Kawachi, I. (2017). Social capital and cognitive decline in the aftermath of a natural disaster: natural experiment from the 2011 Great East Japan Earthquake and Tsunami.

The Lancet Planetary Health, 1(3), e105–e113.

[https://doi.org/10.1016/s2542-5196\(17\)30041-4](https://doi.org/10.1016/s2542-5196(17)30041-4)

Kino, S., Aida, J., Kondo, K., & Kawachi, I. (2020). Long-term trends in mental health

disorders after the 2011 Great East Japan earthquake and tsunami. *JAMA Network Open*, 3(8), e2013437. <https://doi.org/10.1001/jamanetworkopen.2020.13437>

Latest earthquakes. (2005, January 13).

<https://earthquake.usgs.gov/earthquakes/map/?extent=23.60426,112.0166&extent=47>.

Sasaki, Y., Aida, J., Tsuji, T., Koyama, S., Tsuboya, T., Saito, T., Kondo, K., & Kawachi, I.

(2019). Pre-disaster social support is protective for onset of post-disaster depression: Prospective study from the Great East Japan Earthquake & Tsunami. *Scientific Reports*, 9(1), 19427. <https://doi.org/10.1038/s41598-019-55953-7>

Suppasri, A., Koshimura Shunichi, Imai, K., Mas, E., & Imamura, F. (2013). Damage

characteristics and field survey of the 2011 Great East Japan tsunami. *Coastal Engineering Journal*, 55(1), 1350002. <https://doi.org/10.1142/S0578563413500020>

The official report of the Fukushima Nuclear Accident Independent

Investigation Commission. (2012). National Diet of Japan Fukushima Nuclear Accident Independent Investigation Commission.

Tsunamis | National Centers for Environmental Information (NCEI). (n.d.).

[Www.ncei.noaa.gov. https://www.ncei.noaa.gov/products/natural-hazards/tsunamis-earthquakes-volcanoes/tsunamis](https://www.ncei.noaa.gov/products/natural-hazards/tsunamis-earthquakes-volcanoes/tsunamis)

World Nuclear Association. (2025, February 4). *Nuclear Power in Japan- World Nuclear Association*.

[World-Nuclear.org. https://world-nuclear.org/information-library/country-profiles/countries-g-n/japan-nuclear-power](https://world-nuclear.org/information-library/country-profiles/countries-g-n/japan-nuclear-power)